

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('tips')

print("Dataset Shape:", df.shape)
df.head()
```

Dataset Shape: (244, 7)

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   total_bill      244 non-null    float64
1   tip             244 non-null    float64
2   sex             244 non-null    category
3   smoker          244 non-null    category
4   day             244 non-null    category
5   time            244 non-null    category
6   size            244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

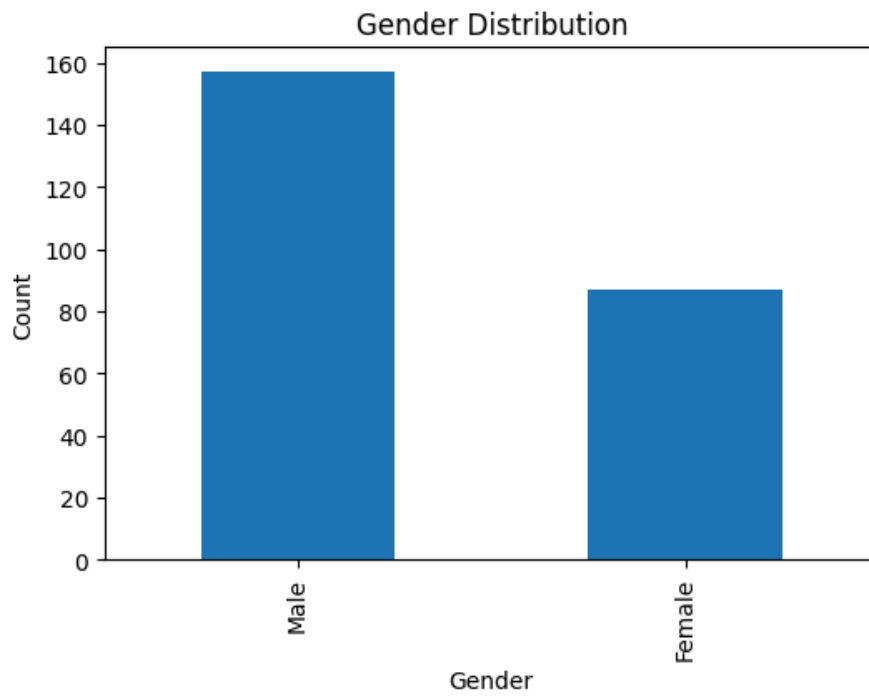
```
df.describe()
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
plt.figure(figsize=(6,4))
df['sex'].value_counts().plot(kind='bar')

plt.title("Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Count")
```

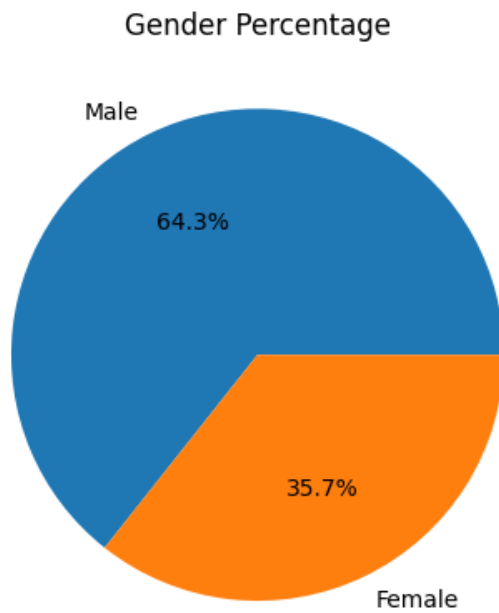
```
plt.show()
```



```
df['sex'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title("Gender Percentage")
plt.ylabel("")

plt.show()
```



Start coding or [generate](#) with AI.

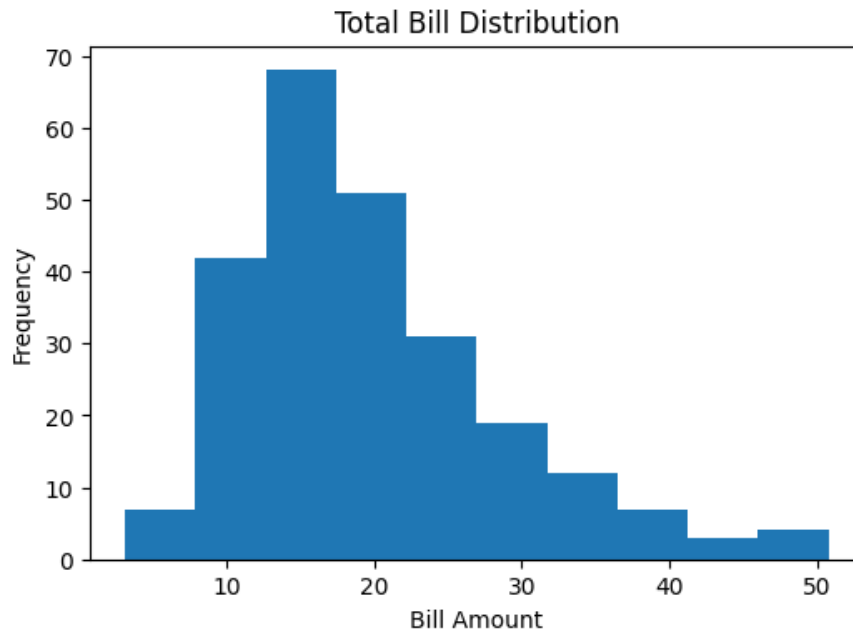
Start coding or [generate](#) with AI.

```
plt.figure(figsize=(6,4))

plt.hist(df['total_bill'], bins=10)

plt.title("Total Bill Distribution")
plt.xlabel("Bill Amount")
plt.ylabel("Frequency")

plt.show()
```

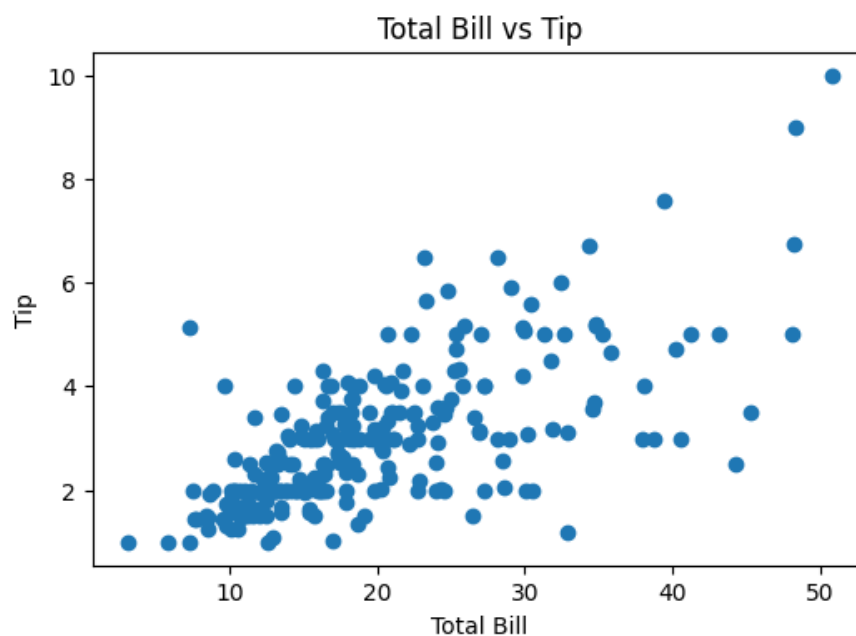


```
plt.figure(figsize=(6,4))

plt.scatter(df['total_bill'], df['tip'])

plt.title("Total Bill vs Tip")
plt.xlabel("Total Bill")
plt.ylabel("Tip")

plt.show()
```

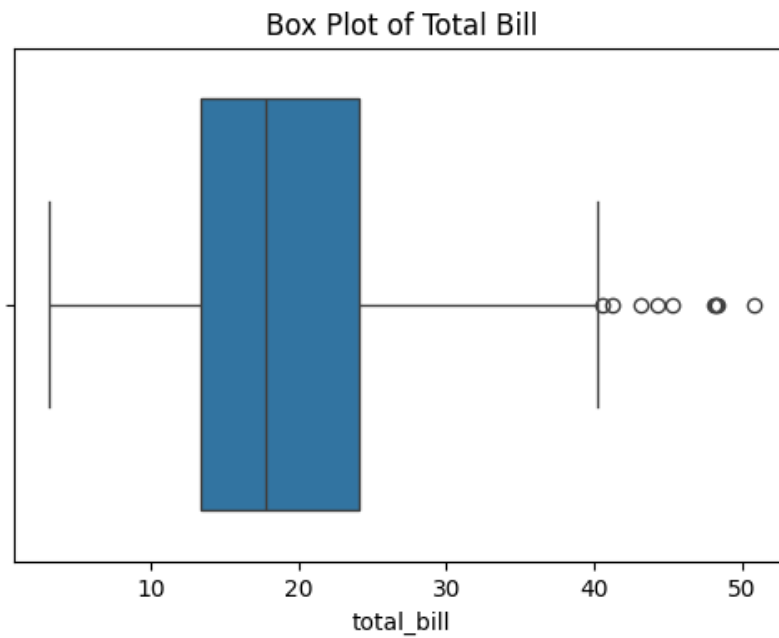


```
plt.figure(figsize=(6,4))

sns.boxplot(x=df['total_bill'])

plt.title("Box Plot of Total Bill")

plt.show()
```

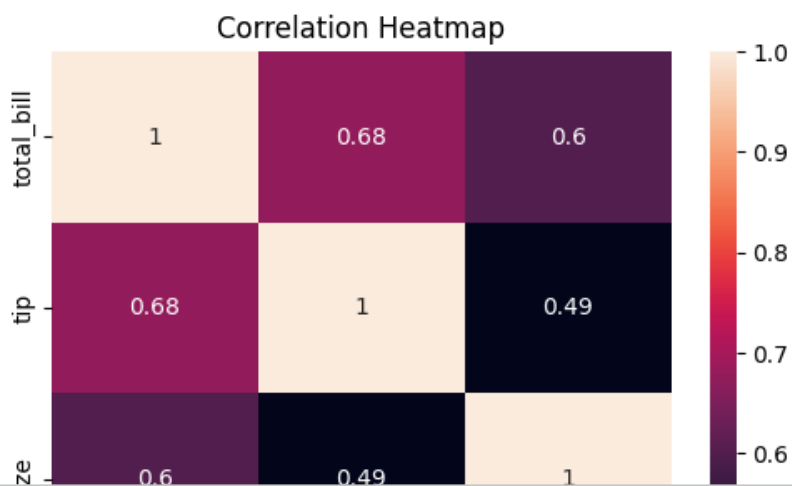


```
plt.figure(figsize=(6,4))

sns.heatmap(
    df.select_dtypes(include='number').corr(),
    annot=True
)

plt.title("Correlation Heatmap")

plt.show()
```



```
fig, ax = plt.subplots(2,2, figsize=(10,8))

df['sex'].value_counts().plot(
    kind='bar',
    ax=ax[0,0]
)

ax[0,0].set_title("Gender Distribution")
```

```
ax[0,1].hist(df['total_bill'])
ax[0,1].set_title("Bill Distribution")
```

```
ax[1,0].scatter(
    df['total_bill'],
    df['tip']
)
ax[1,0].set_title("Bill vs Tip")
```

```
sns.boxplot(
    x=df['total_bill'],
    ax=ax[1,1]
)
ax[1,1].set_title("Box Plot")
```

```
plt.tight_layout()
plt.show()
```

